LARGE LANGUAGE USER INTERFACES: VOICE INTERACTIVE USER INTERFACES POWERED BY LLMS

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ABSTRACT

The evolution of Large Language Models (LLMs) has showcased remarkable capacities for logical reasoning and natural language comprehension. These capabilities can be leveraged in solutions that semantically and textually model complex problems. In this paper, we present our efforts toward constructing a framework that can serve as an intermediary between a user and their user interface (UI), enabling dynamic and real-time interactions. We employ a system that stands upon textual semantic mappings of UI components, in the form of annotations. These mappings are stored, parsed, and scaled in a custom data structure, supplementary to an agent-based prompting backend engine. Employing textual semantic mappings allows each component to not only explain its role to the engine but also provide expectations. By comprehending the needs of both the user and the components, our LLM engine can classify the most appropriate application, extract relevant parameters, and subsequently execute precise predictions of the user's expected actions. Such an integration evolves static user interfaces into highly dynamic and adaptable solutions, introducing a new frontier of intelligent and responsive user experiences.

Keywords Large Language Models, Context-Aware Pervasive Systems, Semantic Modelling, Agent-Based Reasoning, Human-Computer Interaction

1 Introduction

The modern world relies on and is driven by software. Embedded systems, command-line interfaces, web and desktop applications are present across systems all around the world. The predominant way in which we interact with software is through user interfaces. Their ease of use coupled with their intuitive nature has allowed for user interfaces to become an indispensable tool across contemporary software. These systems serve as a visually appealing packaging of function calls and event handlers, allowing complex event pipelines and data flows to be abstracted by buttons, text fields, menus, and so on. The catalyst for our research is grounded in the critical examination of current user interface systems, which, while omnipresent and essential, do not come without limitations. Traditional implementations of UIs often require the user to adhere to the methods and system constraints established by the developer and their design. This means that all UI components in an application have specific triggers and their invocation is limited to a strict sequence of user inputs within the applications' available features.

The substantial advancements made in the space of large language models, especially over the past few years, have exhibited a true “cognitive" potential. This potent ability has unveiled innumerable new opportunities to revolutionize the way our software systems are expected to operate. It became clear to us that an end-to-end architecture employing textual semantic modelling of contemporary user interfaces, as well as leveraging familiar user prompting behaviour, could be crafted. Within our research, we explore our vision and progress toward developing such a UI architectural paradigm, which benefits from the integration of a custom multimodal engine powered by LLMs and state-of-the-art transformer models.

Our proposed framework aims to textually model the semantics of UI components, giving them the ability to communicate effectively with an LLM-driven agent environment. The automated generation of metadescriptors and annotations modelling each UI component allows us to form a data structure that streamlines agent-based parsing and entity extraction. Through this, we facilitate a synergistic interaction between the software and the user, enabling a more intuitive and user-centred experience, rather than requiring the user to conform to the software’s predefined structures. Thus, our framework aids in abstracting monotonous UI interactions with prompting mechanisms that serve as “cognitively aware", powering automated function calling and data flow pipelines. This translates to fully speech-based and intelligent control over visual user interfaces.

The benefits of our paradigm and its extensions are vast. Firstly, it would relieve the user of restricted, stringent and static rulesets when utilizing software. This would fundamentally change the landscape of how applications are developed, used and integrated across all sectors. It would allow for real-time intelligent and efficient control over complex tasks, enabling users to achieve substantially more, in significantly less time. They would be capable of multitasking through speech and executing complicated operations with real-time UI responses. Secondly, such a framework would further democratize software by drastically reducing learning curves, specifically for intricate software. A user would be able to provide their intended actions through speech and textual input and delegate heavy interactions to the framework, letting it reason with the UI to get the job done. Another major benefit of this framework is the streamlining of application development. User interfaces would be simplified substantially, hosting only the most necessary components. Furthermore, developers can generate annotations efficiently or automatically, allowing them to integrate functionality with far fewer input UI components.

The paper is organized in the following manner. In section 2, we showcase related works that introduce what has already been achieved in related domains, as well as where we fill gaps that are present. In section 3, relevant challenges in the development of our framework and how they were overcome are thoroughly explained. Section 4 is where our framework is formally proposed in detail. Section 5 discusses evaluations of the framework’s accuracy and precision, along with showcasing comparisons between the integration of various LLM models. Finally, sections 6 and 7 discuss future directions, final thoughts, and conclude the paper.

2 Related Work

The interest in applying large language models to user-centred domains is only natural, given that the greatest strength of these models is their semantic comprehension and reasoning abilities. Many research endeavours are focused on using LLMs in the context of human-computer interaction, crafting solutions for increased adaptation to user actions and behaviour. This includes training LLMs to better incorporate prompts from users through guided training pipelines which facilitate accurate execution of user commands [1], contextualization embeddings which grant more personalized responses [2], as well as implementing low-code interaction frameworks offering more intuitive control mechanisms over LLM instructing [3]. These works have pushed the capabilities of LLMs forward and opened up more powerful and efficient solutions for user-focused systems. Other efforts are being made in making LLMs increasingly flexible and versatile tools for human-computer interaction, pushing for “journey" and context-aware interactions through few-shot learning methods [4], analysis of user-interest journeys, and desires [5]. Training language models to teach themselves how to use tools in a self-supervised fashion and enabling zero-shot performance through demonstrations of API tools has also broadened the horizons of how LLMs can be integrated into existing pipelines [6].

Many works, similar to those mentioned, are geared towards direct interactions with LLMs, aiming to alleviate reliance on or bypass traditional UIs in favour of enhanced one-on-one sessions between the user and LLM systems. This can significantly improve efficiency and enable more complex operations. However, we find that there is a lack of focus on adapting and evolving contemporary user interfaces. While many system implementations would benefit from the dismissal of UIs, the focus on adding intelligence to their operations is crucial for the next generation of software.

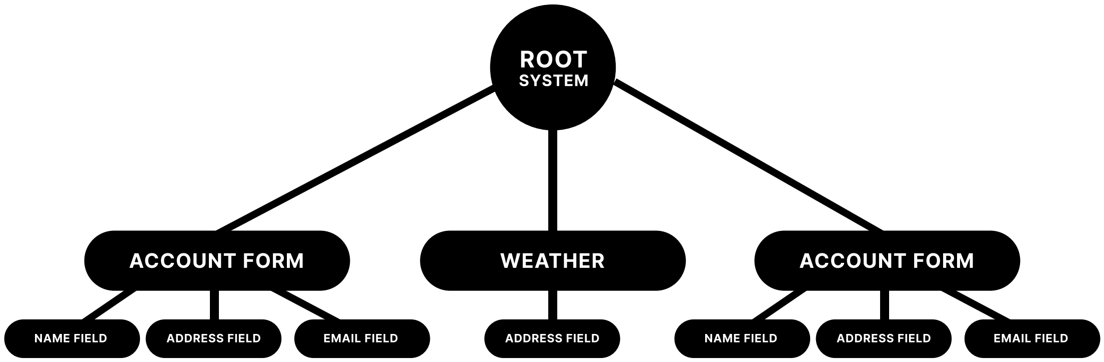

Figure 1: Visual representation of tree data structure used to store application meta descriptions

Users need to be able to visually observe their data as well as the UI's responses to their commands. They must be able to easily provide follow-up instructions with confidence that the framework will adapt in real time, equipping them with the ability to complete operations, simple or complex, with equal convenience. Eliminating learning curves or time-intensive tasks through intelligent speech control is invaluable.

These factors are what led us towards the research and creation of a two-component framework that revolutionizes both ends of the system. A paradigm that changes the way we develop applications, harnessing the power of LLMs to comprehend user prompts and permit back-and-forth agent-based reasoning with our dynamic annotation tree that semantically models the nature of all available UI components. This approach not only facilitates task efficiency, customization, and user-adaptive behaviour, but also simplifies user interface development and consumption, allowing applications to deliver valuable context, user control, and accessibility. We aim to leverage this architecture to make technology more approachable and usable for a broader audience.

3 Challenges

The development of such a framework is not without its challenges. An effective fusion of LLMs and event-driven UIs is a complex task with many critical factors of concern. All classification, processing, parsing, and model inference must be achieved in real-time and the architecture must be capable of scaling and integrating unseen sub-applications in zero-to-few shots. Additionally, the framework's interpretation accuracy of ambiguous user inputs must be top-notch. The greatest challenges encountered during our research and implementation of this framework are discussed in this section.

3.1 Integration Complexity

The underlying challenge across the entire research endeavour was effectively fusing LLMs with event-driven UIs. Developing an effective framework capable of being scaled with increasing application counts, as well as fostering potential for implementation as an operating system was always a priority. A paradigm as dynamic as the software it intends to power was vital. Implementing a brute-force hard-coded solution would not serve a great purpose or carry the potential for scalability and impact required for next-generation software. This is why we took our time and carefully crafted each component of this framework with scaling in mind, leaving room for future advanced integrations, such as more powerful LLM model configurations and automated agent fine-tuning.

3.2 Annotation Data Structure Selection & Traversal

Another critical concern was selecting a data structure to effectively represent the system, its applications, and all UI components. Through our research, a tree structure made for the most accurate representation of the overall system, whilst also allowing for efficient traversal and information retrieval, vital for real-time responses further on.

The specific implementation of this tree structure still poses room for iterative research concerning scalability, however, regardless of the implementation method, the need for top-down level-by-level comparison is where the tree structure